

Group 7 Project - Milestone 1 (Due Monday, June 9)

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2. Application Description

Our app will help users to make informed decisions about the best places to live depending on the conditions such as social services, health stats, economic outlook, job prospect, crime etc. Our target audience are home buyers, students, families, remote workers, real estate agents. It is inspired by platforms like [Niche](#), [AreaVibes](#), [SpotCrime](#), and [City-Data](#).

3. Dataset Background

- **Realtor.com Historical Data:**
 - It provides historical data on the US housing market per zipcode. The data ranges from 2016 to 2025. We can make comparison such as how does this city/zipcode compared to national average
 - **Attribute:** median_listing_price, number of active listing, new listing count, zipname (4 attributes)
 - Total size: 2.9 million entries of data before preprocessing
- ZIP:**
- Count: 2.923×10^6
 - Mean: 49,000
 - Std Dev: 27,681
 - Min: 1,001
 - 25%: 26,201
 - 50% (Median): 48,740
 - 75%: 72,044
 - Max: 99,921
- Median_listing_price:**
- Count: 2.923×10^6
 - Mean: 377,144

- Std Dev: 793,594
- Min: 1
- 25%: 157450
- 50% (Median): 267000
- 75%: 429900
- Max: 279,000,000

Active_listing_count:

- Count: 2.923×10^6
- Mean: 32.5
- Std Dev: 59
- Min: 0
- 25%: 3
- 50% (Median): 11
- 75%: 38
- Max: 2646

- **CDC Places, Local Data for Better Health (ZCTA Data 2024 release):**

- This 2021-2022 dataset provides exactly 40 measures of health in terms of the frequencies of their occurrence within the populations of exactly 32520 US ZTCAs (i.e., zip code tabulation areas, which are a proxy for residential ZIP codes)
- Total size: ~250MB CSV containing 1.2 million entries of data before preprocessing
 - **Note:** We plan to use only 4 of the 40 measures, which translates to roughly 130,000 entries postprocessing
- Attribute: Year, LocationName, DataSource, Category, Measure, Data_Value_Unit, Data_Value, Data_Value_Footnote_Symbol, Data_Value_Footnote, Low_Confidence_Limit, High_Confident_Limit, Totalpop18plus, TotalPopulation, Geolocation, LocationID, CategoryID, MeasureId, DataValueTypeID, Short_Question_Text
- Summary statistics:
 - Data_Value (in %)
 - Mean 26.3
 - Min 0.2
 - 25th percentile 9.7
 - 50th percentile (median) 17.2
 - 75th percentile 34.7
 - Max 97.6
 - TotalPopulation
 - Mean 10552
 - Median 2933
 - Min 52
 - Max 135256

- **US Fire Administration:**

It provides the number of fire departments and activer personnel per zip code.

Attributes: zipcode, total_fire_departments, total_number_of_stations, total_active_firefighters

Total size: 26k entries of data before preprocessing

Summary statistics:

total_fire_departments

- Count: 20,438
- Mean: 1.325
- Std Dev: 0.776
- Min: 1
- 25%: 1
- 50% (Median): 1
- 75%: 1
- Max: 15

total_number_of_stations

- Count: 20,438
- Mean: 2.570
- Std Dev: 4.927
- Min: 0
- 25%: 1
- 50% (Median): 1
- 75%: 3
- Max: 268

total_active_firefighters

- Count: 20,438
- Mean: 52.230
- Std Dev: 133.120
- Min: 0
- 25%: 20
- 50% (Median): 30
- 75%: 52
- Max: 10,482

- [Homeland Infrastructure Foundation-Level Database:](#)

- [Police](#): number of police stations and active personnel per zipcode.
Attributes: zipcode, total_police_departments, number_fulltime_police_officers, number_part_time_police_officers, total_number_of_police_officers.
Total size: 23k entries of data before preprocessing.

Summary statistics:

total_police_departments

- Count: 15,398

- Mean: 1.525
- Std Dev: 0.952
- Min: 1
- 25%: 1
- 50% (Median): 1
- 75%: 2
- Max: 19

number_fulltime_police_officers

- Count: 15,398
- Mean: 44.477
- Std Dev: 377.336
- Min: 0
- 25%: 1
- 50% (Median): 7
- 75%: 28
- Max: 36,096

number_part_time_police_officers

- Count: 15,398
- Mean: 2.639
- Std Dev: 8.264
- Min: 0
- 25%: 0
- 50% (Median): 0
- 75%: 3
- Max: 486

total_number_of_police_officers

- Count: 15,398
- Mean: 47.116
- Std Dev: 377.871
- Min: 0
- 25%: 2
- 50% (Median): 10
- 75%: 31
- Max: 36,096

- [Hospitals](#): number of hospitals per zipcode.

Attributes: zipcode, count.

Total size: 8.3k entries of data.

Summary statistics:

ZIP:

- Count: 18,987

- Mean: 48,624.6
- Std Dev: 28,478.2
- Min: 602
- 25%: 26,054.5
- 50% (Median): 46,932
- 75%: 73,723
- Max: 99,921

Count:

- Count: 18,987
- Mean: 6.964
- Std Dev: 8.1
- Min: 1
- 25%: 1
- 50% (Median): 4
- 75%: 9.5
- Max: 102

- [ChildCare](#): number of childcare centers per zipcode.
Attributes: zipcode, count.
Total size: 132k entries of data before preprocessing.
Summary statistics:

ZIP

- Count: 18,987
- Mean: 48,624.601
- Std Dev: 28,478.225
- Min: 602
- 25%: 26,054.5
- 50% (Median): 46,932
- 75%: 73,723
- Max: 99,921

Count

- Count: 18,987
- Mean: 6.964
- Std Dev: 8.188
- Min: 1
- 25%: 1
- 50% (Median): 4
- 75%: 9.5
- Max: 102

- [United States ZipCode.org](#):
 - It provides city, state, county name per zipcode. The dataset also provides latitude and longitude data.

- Attribute: primary_city, county, state, latitude, longitude (5 attributes)
 - Total size: 42736 entries of data before preprocessing
 - [US Census Data AVG Income:](#)
 - This provides zip code level data on Household income. It goes further into details about income level by each population group. The Geographic Area name has a few preceding text "ZTCA5 " that will need to be stripped from the column
 - Attribute: Geographic Area Name, Mean Income(2 attributes)
 - Total size: 33774 entries of data before preprocessing. Among those, 1854 rows are null/ empty value. We have 31919 rows for data by zip code after removing all null from the list
- Mean Income**
- Count: 31919
 - Mean: 96684
 - Std Dev: 45046.7
 - Min: 4280
 - 25%: 70605
 - 50% (Median): 86630
 - 75%: 110068
 - Max: 1210208

4. Example Query

1. Which zipcodes have the highest combined number of hospitals, police departments, childcare centers, and fire departments?
2. What is the average number of childcare centers in zipcodes with at least 1 hospital, and 5 police departments, but no fire departments?
3. Which zip code within this city has the lowest price for median home price? This would required CTE to determine the zip code in the same city and aggregation to determine the minimum value?
4. Determine the home price growth of this zip code for the past 5 years
5. Find ZIP codes that are underserved in terms of public safety and health by identifying the ZIP codes with (a) low public safety/health staffing and (2) high rates of poor health outcomes
6. Find ZIP codes with the greatest level of growth as measured by increase in home sales, improvements in health outcomes, and increase in number of public safety/health facilities
7. Average distance to Nearest hospital by price tier
8. Rank cities by safety to sale ratio
9. Find a list of affordable zip code for user by selecting state. Affordability is calculated by a ratio of home price and income. It will require joining the income table to home price data and use aggregation in order to determine a list of ratio less than or equal to 3.
10. Find zip code that are underserved in terms of healthcare resource by joining CDC dataset for population and hospital data set for number of hospital in order to group by zip code to calculate the ratio of population to number of hospital

